

Advanced (Habanero)

- Q1.** What is the vertex of the parabola $y = x^2 + 4x + 4$?
(A) $(-2, 0)$
(B) $(-2, 4)$
(C) $(0, 4)$
(D) $(2, 0)$
(E) $(2, 4)$
- Q2.** Which of the following is the axis of symmetry for $y = x^2 - 6x + 8$?
(A) $x = -2$
(B) $x = -1$
(C) $x = 2$
(D) $x = 3$
(E) $x = 4$
- Q3.** What are the x -intercepts of $y = x^2 + 2x - 3$?
(A) $(-3, 0)$ and $(1, 0)$
(B) $(-2, 0)$ and $(1, 0)$
(C) $(-1, 0)$ and $(3, 0)$
(D) $(-1, 0)$ and $(2, 0)$
(E) $(-2, 0)$ and $(3, 0)$
- Q4.** For $y = -x^2 + 4x - 3$, what is the y -intercept?
(A) $(-3, 0)$
(B) $(0, -3)$
(C) $(0, 3)$
(D) $(1, 0)$
(E) $(3, 0)$
- Q5.** What is the maximum value of $y = -x^2 + 2x + 3$?
(A) -1
(B) 0
(C) 2
(D) 4
(E) 6
- Q6.** The direction of the parabola $y = 2x^2 - 4x - 6$ is
(A) Down
(B) Up
(C) Left
(D) Right
(E) Horizontal
- Q7.** The vertex form of $y = x^2 + 4x + 4$ is
(A) $y = (x - 2)^2$
(B) $y = (x + 2)^2$
(C) $y = (x - 1)^2 + 1$
(D) $y = (x + 1)^2 - 1$
(E) $y = (x - 2)^2 + 4$
- Q8.** For $y = x^2 - 2x - 3$, the axis of symmetry is
(A) $x = -1$
(B) $x = 0$
(C) $x = 1$
(D) $x = 2$
(E) $x = 3$

Open Ended Questions (FRQ)

Q9. Graph $y = x^2 - 4x - 3$ and identify its vertex, axis of symmetry, x -intercepts, y -intercept, and direction.

Q10. Find the maximum or minimum value of $y = -x^2 + 2x + 1$. Justify your answer with the graph and calculations.

Chef's Tips

- For questions 1-8, select the correct answer from the given options.
- For questions 9-10, use complete sentences and provide clear, concise answers.
- Show all work and calculations for questions 9-10.